

FICHER CONFIGURATION MQTT TLS

Architecture :

Raspberry Pi

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| MQTT TLS : port 8883

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Mosquitto Broker

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Backend API

1. Installation Mosquitto

Linux :

```
sudo apt update
```

```
sudo apt install mosquitto mosquitto-clients
```

2. Configuration Broker MQTT

Fichier :

```
/etc/mosquitto/mosquitto.conf
```

Contenu :

```
# Port MQTT sécurisé  
listener 8883
```

```
# TLS  
cafile /etc/mosquitto/certs/ca.crt  
  
certfile /etc/mosquitto/certs/server.crt  
  
keyfile /etc/mosquitto/certs/server.key
```

```
# Désactiver accès anonyme

allow_anonymous false

# Authentification

password_file /etc/mosquitto/passwd

# Sécurité

require_certificate true

tls_version tlsv1.3
```

3. Création certificat CA

Créer dossier :

```
mkdir certs

cd certs
```

Créer autorité :

```
openssl genrsa -out ca.key 4096
```

Créer certificat :

```
openssl req \
-x509 \
-new \
-key ca.key \
-out ca.crt \
-days 365
```

4. Certificat serveur MQTT

Créer clé :

```
openssl genrsa \
-out server.key 2048
```

Créer certificat :

```
openssl req \
-new \
-key server.key \
-out server.csr
```

Signer :

```
openssl x509 \
-req \
-in server.csr \
-CA ca.crt \
-CAkey ca.key \
-out server.crt \
-days 365
```

5. Création utilisateur MQTT

Créer utilisateur :

```
sudo mosquitto_passwd \
-c /etc/mosquitto/passwd raspberry
```

6. Redémarrage service

```
sudo systemctl restart mosquitto
```

Vérifier :

```
systemctl status mosquitto
```

7. Test connexion sécurisée

Publisher (Raspberry Pi)

```
mosquitto_pub \
-h serveur.com \
-p 8883 \
--cafile ca.crt \
--cert client.crt \
--key client.key \
-t camera/detection \
-m "face_detected"
```

Subscriber (Backend)

```
mosquitto_sub \
-h serveur.com \
-p 8883 \
--cafile ca.crt \
--cert client.crt \
--key client.key \
-t camera/detection
```

8. Topics MQTT du projet

Organisation :

camera/detection

camera/status

security/alert

iot/heartbeat

Exemple message :

```
{  
  "camera": "cam01",  
  "result": "unknown_face",  
  "score": 0.92,  
  "time": "2026-06-18"  
}
```